

Lakshmi Manasa Borra

AI Engineer

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PROFESSIONAL SUMMARY

AI and Machine Learning Engineer with nearly 4 years of experience in building production-grade AI applications, agentic AI solutions, scalable data platforms, and cloud-native systems using Python. Hands-on experience across the full AI lifecycle, including data engineering, model development, deployment, monitoring, and optimization. Developed REST APIs, retrieval-augmented AI systems, intelligent recommendation platforms, care management solutions, and model-serving infrastructure using FastAPI, Docker, AWS, and modern MLOps practices. Experienced in leveraging LLMs, semantic search, agentic workflows, and machine learning models to solve complex business challenges across finance, healthcare, and education domains. Strong collaborator with a proven ability to quickly learn new technologies and deliver scalable solutions in cross-functional environments.

TECHNICAL SKILLS

Programming Languages: Python, SQL, R, PySpark

Backend & Software Engineering: FastAPI, REST APIs, API Integration, Microservices Concepts, Distributed Systems Fundamentals

Databases: PostgreSQL, MySQL, Snowflake, Vector Databases, SQL Query Optimization

Machine Learning & AI: PyTorch, TensorFlow, Scikit-Learn, XGBoost, LightGBM, CNNs, RNNs, Transformers, BERT, Forecasting, Recommendation Systems

Generative AI & Agentic Systems: LangChain, LangGraph, LlamaIndex, RAG Pipelines, FAISS, Semantic Search, Agentic Workflows, Prompt Engineering, Tool Calling, LLM Evaluation

Cloud, MLOps & Platform Engineering: AWS (SageMaker, EC2, S3), GCP, Azure, Docker, Kubernetes, MLflow, CI/CD, Model Deployment, Model Monitoring

Development Tools: Git, GitHub, GitHub Copilot, ChatGPT, Windsurf

Data Engineering & DataOps: Airflow, ETL/ELT Pipelines, Distributed Data Processing, Workflow Orchestration, Data Quality Monitoring, Batch Processing, Data Modeling

PROFESSIONAL EXPERIENCE

Agentic AI Engineer | JPMorgan Chase & Co. | Remote

July 2025 – Present

- Engineered agentic AI solutions integrating LLMs, retrieval pipelines, and reasoning workflows for financial operations.
- Designed ML pipelines using Python, SQL, Airflow, and MLflow, improving reliability and deployment efficiency.
- Developed REST APIs and model services using FastAPI, SageMaker, EC2, and S3.
- Deployed containerized AI applications using Docker, improving portability and deployment consistency.
- Implemented automated validation and monitoring frameworks ensuring data integrity and model performance.

AI Engineer | UnitedHealth Group | Hyderabad, IND

May 2023 – Dec. 2023

- Developed healthcare AI solutions integrating semantic search, vector retrieval, and LLMs for care management.
- Designed scalable retrieval workflows integrating SQL, PostgreSQL, FastAPI, and distributed search architectures.
- Developed feature engineering pipelines improving model accuracy, data consistency, and system reliability.
- Supported cloud deployment and performance monitoring across machine learning environments.
- Conducted benchmarking and model evaluation for large-scale retrieval and AI systems.

Machine Learning Engineer | UpGrad INSOFE | Hyderabad, IND

Oct 2021 – Apr 2023

- Built machine learning workflows integrating recommendation engines, predictive analytics, and NLP for personalized learning.
- Developed recommendation systems, predictive models, and NLP applications using PySpark, Python, BERT.
- Developed scalable data pipelines supporting large-scale datasets and production AI applications.

Machine Learning Engineer | HCL | Hyderabad, IND

Mar 2021 – Sep. 2021

- Developed NLP, recommendation, and deep learning models using TensorFlow, PyTorch, Keras, CNNs, RNNs, LSTMs, and Transformers.
- Built Intelligent Patient Risk and Clinical Sentiment Analysis Platform using NLP, computer vision, psychometric modeling.
- Engineered production-ready AI/ML pipelines, applied transfer learning, created custom datasets, communicated stakeholder insights effectively

NOTABLE PROJECTS

- Intelligent Financial Operations Copilot** – LLMs · Agentic AI · RAG · AWS
Engineered agentic AI solutions integrating LLMs, retrieval pipelines, and reasoning workflows for financial operations.
- AI-Driven Care Management Assistant** – FAISS · Semantic Search · LLMs · FastAPI
Developed healthcare AI solutions integrating semantic search, vector retrieval, and LLMs for care management.
- Intelligent Student Learning & Recommendation Platform** – BERT · Recommendation Systems · PySpark · NLP
Built machine learning workflows integrating recommendation engines, predictive analytics, and NLP for personalized learning.

EDUCATION

M.S. in Data Science | New Jersey Institute of Technology

Jan 2024 – May 2025

PG Certificate in Computational Data Science | INSOFE & Case Western Reserve University

Nov 2021 – Oct 2022

B.Tech. in Mechanical Engineering | Geethanjali College of Engineering & Technology

Jul 2018 – Jul 2022

VOLUNTEERING ASSOCIATION

Project: Multi-Agent Orchestrated Retrieval-Augmented Generation (RAG) System for Academic Knowledge Management

- Designed a **multi-agent RAG platform** using **Azure OpenAI, AutoGen, and LangChain** for academic knowledge retrieval.
- Built OCR, preprocessing, embedding, and hybrid search pipelines using **Azure Cognitive Search and FAISS**.
- Deployed a secure React-Flask application on AKS with SSO, CI/CD, monitoring, and citation validation.